

Online Appendix:

“Supply Chain Risk: Changes in Supplier Composition and Vertical Integration”

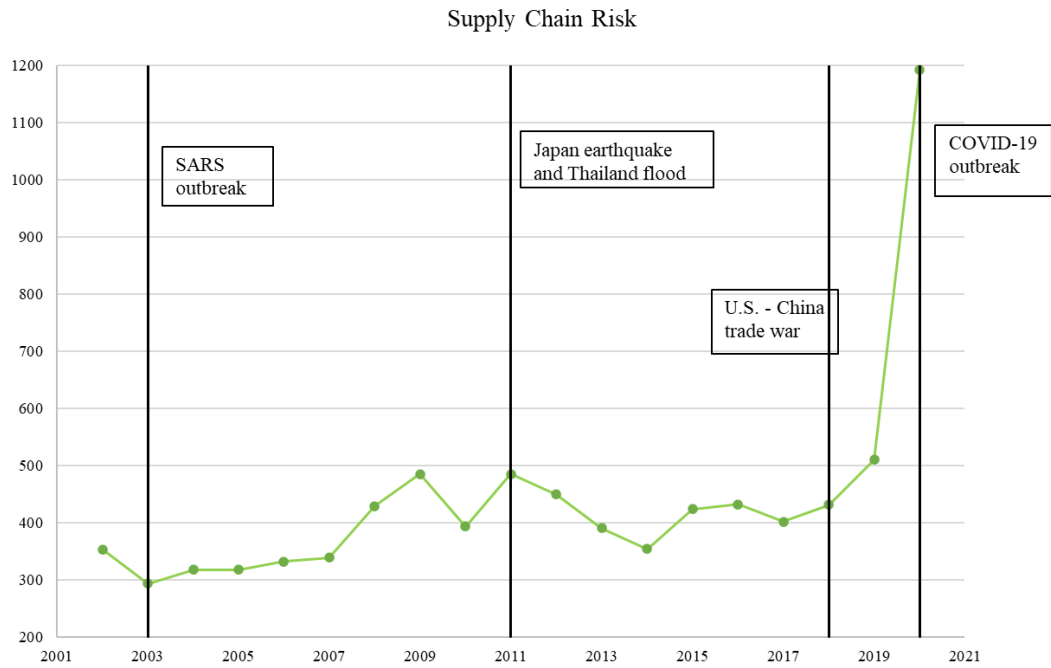
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Appendix A Variable definitions.

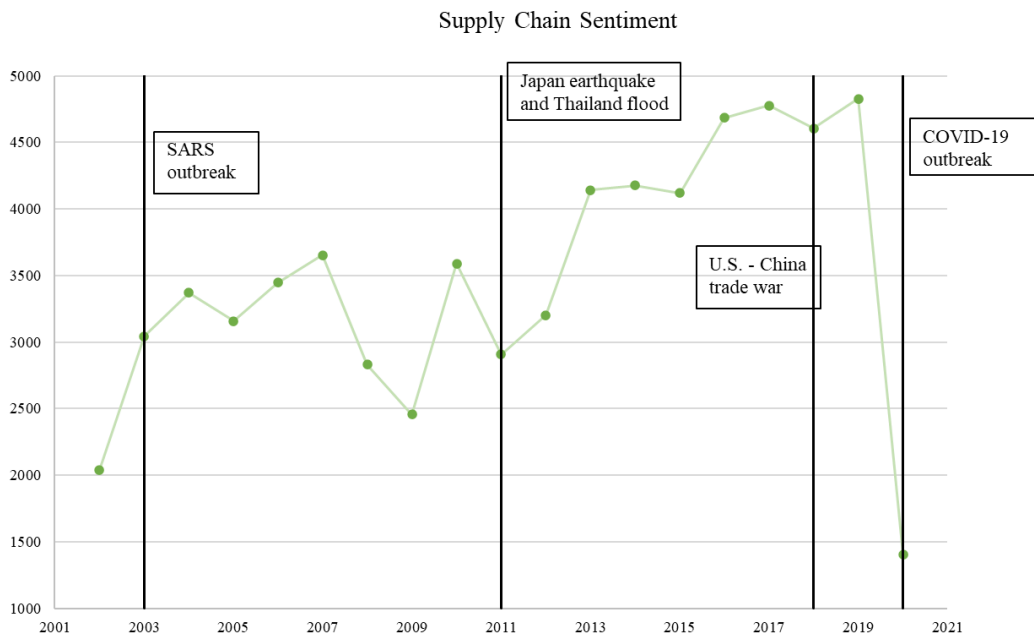
Variables	Definition
SCRisk	Firm-level supply chain risk measure constructed from earnings calls
SCSentiment	Firm-level supply chain sentiment measure constructed from earnings calls
Suppliers' SCRisk	The maximum SCRisk of a firm's suppliers
Different continents	Fraction of a firm's suppliers located in a continent different from that of the firm over the total number of suppliers
Relative size	Focal firm's total assets scaled by its suppliers' average total assets
Average number of suppliers by industry	The average of a firm's number of suppliers by input industry
Multinational	An indicator variable that equals one if the firm has geographical segments in other countries based on Compustat segment data.
Customer dependence	An indicator variable that equals one if the firm reports major customers that purchase at least 10% of its sales in Compustat segment data
Market share	Firm's sales scaled by the 2-digit SIC industry's total sales
Financial constraint	A dummy variable that equals one if the firm's Whited-Wu (2006) measure for financial constraints is above the sample median
Institutional ownership	Fraction of shares owned by financial institutions
Size	Natural logarithm of total assets
Tobin's Q	Assets minus cash and cash equivalent securities plus book value on equity scaled by assets
Cash holdings	Cash and cash equivalent securities scaled by total assets
Cash flow	Operating cash flow scaled by total assets
Number of suppliers	A firm's total number of suppliers
Number of suppliers in the same continent	The total number of suppliers in the same continent as the firm
Number of U.S. suppliers	A firm's total number of suppliers in the U.S.
Number of suppliers in different continents	The total number of suppliers in different continents as the firm
Number of industry leader suppliers	A firm's total number of suppliers with sales above the median of their 3-digit SIC industry
M&A with supplier	A dummy variable that equals one if the firm conducts an M&A with a firm from an upstream industry according to the Bureau of Economic Analysis's (BEA) Input-Output tables.
M&A with customer	A dummy variable that equals one if the firm conducts an M&A with a firm from a downstream industry according to the Bureau of Economic Analysis's (BEA) Input-Output tables.
Unrelated M&As	A dummy variable that equals one if the firm conducts an M&A with a firm that is not in an upstream, downstream, or the same industry.

Figure IA.1 Supply Chain Risk and Sentiment over Time.

Panel A. SCRisk

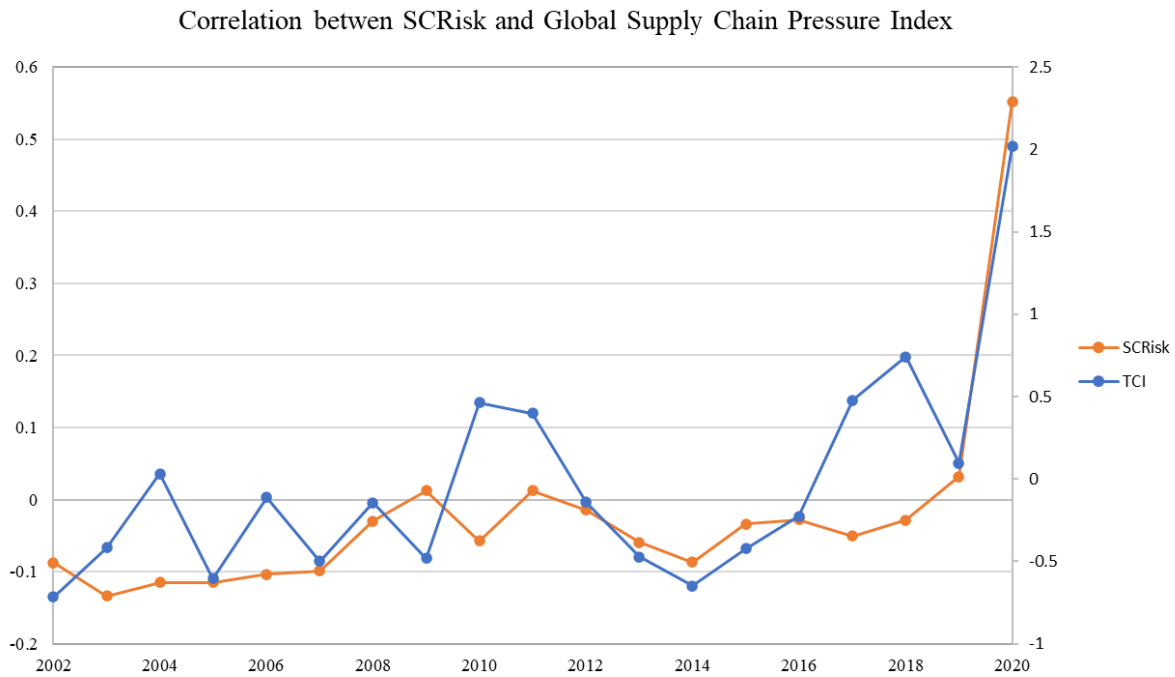


Panel B. SCSentiment



Notes: This figure shows the mean of SCRisk and SCSentiment along with indicators for key events related to supply chain shocks. SCRisk and SCSentiment are scaled up by a factor of 1,000.

Figure IA.2 Supply Chain Risk and Transportation Costs



Notes: This figure shows the mean of SCRisk and the global transportation costs index (TCI) developed by Benigno et al. (2022), both measures are standardized by subtracting the sample mean and dividing by the sample standard deviation.

Table IA.1 Top 100 Bigrams and Their Weights.

Bigram	Weight	Bigram	Weight	Bigram	Weight
supply_chain	761.63	strategic_fit	32.64	of_capacity	18.28
the_supply	281.15	of_safety	32.21	chain_and	18.28
a_supply	146.23	chain_to	31.77	demand_the	18.28
the_retailer	133.18	to_order	31.34	production_cost	18.28
of_demand	104.89	of_scale	31.34	the_peak	17.41
the_manufacturer	104.89	supply_and	30.03	variable_cost	16.97
lead_time	98.79	if_demand	30.03	the_cycle	16.54
demand_is	93.14	the_aggregate	29.59	the_safety	16.54
of_product	79.21	the_lot	29.16	to_supply	16.54
the_demand	74.42	chain_is	28.72	demand_to	16.54
the_supplier	71.81	customer_order	27.42	chain_the	16.54
transportation_cost	64.41	spot_market	26.98	fixed_costs	15.23
of_supply	56.58	time_is	26.55	from_supplier	15.23
transportation_costs	56.58	demand_from	24.81	the_network	14.80
an_order	55.71	the_lead	23.94	million_units	14.80
expected_profit	53.53	lead_times	23.94	a_forecast	14.36
demand_and	52.66	stages_of	23.07	demand_per	14.36
third_party	52.66	the_plant	23.07	uncertainty_and	13.93
supply_chains	52.66	and_supply	23.07	of_orders	13.93
fixed_cost	46.57	the_spot	23.07	a_demand	13.93
the_season	45.26	distribution_network	22.63	the_capacity	13.49
the_quantity	44.83	demand_at	22.63	lower_price	13.49
demand_in	40.91	demand_forecast	22.63	demand_across	13.49
and_demand	39.60	when_demand	22.63	chain_for	13.49
of_transportation	38.73	in_supply	20.46	capacity_to	13.06
revenue_management	38.73	capacity_is	20.46	product_to	13.06
chain_management	38.30	demand_of	20.46	production_capacity	13.06
response_time	37.43	and_transportation	20.02	revenue_sharing	13.06
demand_uncertainty	36.99	supplier_to	20.02	future_demand	12.62
service_level	36.56	customer_orders	19.15	in_transportation	12.62
the_forecast	36.12	global_supply	18.71	product_in	12.19
customer_demand	33.95	high_demand	18.71	square_feet	12.19
economies_of	33.51	to_forecast	18.28	level_is	12.19
order_is	33.08				

Notes: This table reports the 100 bigrams with the highest frequency in the training library used for the construction of SCRisk. The weight column reports the weight of the bigram in the training library.

Table IA.2 Synonyms of Risk Words.

Synonyms of risk words			
ambivalence	fear	niggle	treacherous
ambivalent	fickleness	oscillating	tricky
apprehension	fitful	parlous	uncertain
bet	fitfulness	pending	uncertainties
chance	fluctuant	peril	uncertainty
chanciness	fluctuating	perilous	unclear
chancy	gamble	perilousness	unconfident
changeability	gnarly	possibility	undecided
changeable	hairy	precarious	undependable
changeableness	halting	precariousness	undetermined
changeful	hazard	probability	unforeseeable
chariness	hazardous	prospect	unknown
danger	hazy	qualm	unpredictability
dangerous	hesitancy	quandary	unpredictable
debatable	hesitant	queries	unreliability
defenseless	hesitating	query	unreliable
dicey	iffy	reservation	unresolved
diffidence	imperil	risk	unsafe
diffident	incalculable	risked	unsettled
dilemma	incertitude	riskier	unstable
disquiet	inconstancy	riskiest	unsure
disquietude	indecision	riskiness	unsureness
dodgy	indecisive	risking	untrustworthy
doubt	insecure	risks	vacillating
doubtful	insecurity	risky	vacillation
doubtfulness	instability	scruple	vague
dubiety	irregular	skepticism	vagueness
dubious	irresolute	speculative	variability
endanger	irresolution	sticky	variable
equivocating	jeopardize	suspicion	varying
equivocation	jeopardy	tentative	venture
erratic	likelihood	tentativeness	wager
exposed	menace	threat	wariness
faltering	misgiving	torn	wavering

Notes: This table reports all synonyms of “risk,” “risky,” “uncertain,” and “uncertainty” used to construct SCRisk. Oxford Dictionary is used to identify the synonyms following Hassan et al. (2019).

Table IA.3 Excerpts From Earnings Calls.

Firm Name	Date of Report	Text
Panel A: Excerpts based on SCRisk		
Mercury Systems, Inc.	April 28, 2020	The key supply chain issues that we're facing are twofold. The first is that suppliers may be financially vulnerable. This applies more so to those suppliers that are heavily exposed to the commercial aerospace sector. As you know, commercial aerospace has been significantly more impacted by COVID than defense. The other major supply chain risk is the potential for COVID-related manufacturing disruptions , that is temporary site shutdowns that could affect the supply of U.S. sourced components to Mercury. We're also facing other operational risks , the first being the potential for COVID-related disruptions within Mercury's own manufacturing facilities...That said, the risk does remain elevated.
Select Interior Concepts, Inc.	November 05, 2020	As we look at international supply chain, it's fairly fragmented. And you have considerable risk with respect to tariffs, supply chain , work stoppages at ports, those kinds of things.
NeoPhotonics Corp	April 30, 2020	While we believe there is immediate demand to increase network bandwidth capacity to handle the increased traffic, we continue to see supply chain risks . We have included approximately \$10 million of impacts to Q2 revenue in our outlook due to concerns about supplier shutdowns as they comply with their local public health orders. We expect the supply chain risks to continue into the second half of the year.
SBE, Inc.	May 2006, 2005	Our customers don't provide much forecast visibility resulting in hesitancy throughout the supply chain .
Science Applications International Corp	December 08, 2016	The biggest variability this quarter, and in our portfolio as a whole, is in the supply chain and materials business.
Insteel Industries, Inc.	July 19, 2018	... uncertainty surrounding the availability of our primary raw material , hot-rolled steel wire rod, resulted in speculative purchasing throughout the supply chain and sharp price increases reflecting the 25% tariff that was eventually applied to practically all imports of carbon steel products.

Entegris, Inc	April 26, 2016	As I was mentioning in my prepared remarks, we are seeing increased level of complexity, increased risk of contamination of critical materials in the supply chain at the leading-edge.
IEC Electronics Corp	May 09, 2018	This brings me to another topic: the ongoing global supply chain component constraints . As you know, in fiscal 2018 Q1, we mentioned that one of our challenges, which is affecting the entire industry, was associated with difficult in producing -- in procuring certain electronic components and in some cases, facing long lead times or allocation restrictions due to limited global supplies . These shortages can impact our ability to fulfill our customers' orders and lengthen production times as well as add some amount of unpredictability as we wait for a specific component to complete a job.

Panel B: Excerpts based on SCSentiment

<i>Positive sentiment</i>		
LightInTheBox Holding Co., Ltd.	April 21, 2016	Stronger than expected holiday sales were primarily a result of our improved supply chain management and the stronger support of our suppliers with discounts and sourcing prices for the holiday season.
Coty Inc.	November 9, 2020	Third, our focus and enhancements over the last year to our supply chain continue to support our growth while allowing us to successfully navigate the COVID-19 pandemic. We added another co-packer to our network during the third quarter, providing further capacity, flexibility and the ability to service our customers.
Vitria Technology, Inc.	October 24, 2006	Second, one of our customers, MasterBrands won "Network World's" 2006 Enterprise All-Star Award for its innovative supply chain management application. Powered primarily by Vitria's BusinessWare products, MasterBrands was able to dramatically speed order fulfillment, provide exceptional handling and enable visibility across the supply chain , earning the company an Enterprise All-Star Award
Tuniu Corporation	August 28, 2019	In the travel industry, the supply chain is the vital component that connects the supply and demand. We made strong progress during the last few years in further strengthening our supply chain . We continue to consolidate procurement across the company in order to maximize our bargaining

power with **suppliers** and minimize risk, while better sharing inventory across our various business units.

HanesBrands, Inc.	July 23, 2014	Q2 was another great quarter for Hanesbrands. We expanded operating margins by 210 basis points and grew our earnings 44% to \$1.71 per share, providing further evidence that when you combine our Innovate-to-Elevate strategy, our self-owned supply chain , and strategic acquisitions, we have a great formula for creating shareholder value .
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Newell Brands Inc.	May 1, 2020	<i>Negative sentiment</i> In the month of April, the supply chain disruptions, the retail closures and the consumer purchase pattern shifts contributed to an estimated sales decline in the 25% sales range, which has informed our call out for a challenging second quarter.
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Reed's, Inc.	November 7, 2018	First quarter revenues declined 7.7% on a like-for-like basis, as we encountered temporary supply chain headwinds . Alongside the previously flagged supply chain issues affecting Consumer Beauty and Professional Beauty, our Luxury division was also impacted in Q1 by a disruption in European warehouse, by the U.S. hurricane and by component shortages at certain external suppliers .
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Micrel Semiconductor, Inc.	April 21, 2011	Consistent with our pre-announcement of preliminary first quarter results on April 11, our top line was impacted by an unanticipated reduction in sales to a Korean wireless handset and consumer electronic device manufacturer which moderated product deliveries during the quarter to control inventory levels. The Company also experienced a reduction in overall demand toward the end of the quarter related to disruptions in the worldwide electronics supply chain as a result of the earthquake and tsunami in Japan in March.
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Corning, Inc.	January 27, 2009	The supply chain actually reacted faster and more severely than we anticipated in quarter four. We had thought the supply chain would reduce 75 million square feet in Q4 with the risk of the additional 50. The reality turned out to be far bigger. Supply chain reduced 230 million square feet of glass in quarter for. When you combine the retail weakness and supply chain contraction , you will understand why our glass volumes were down so much in Q4. And by the way, these numbers I'm discussing here were for the entire industry. So it was a difficult and disappointing quarter.
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Ocean freight, Inc.	May 27, 2011	Let's shift gears now and look at the recent developments in the dry bulk market . The market for the first quarter of 2011
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resulted in a **very low** freight rate environment, in many cases, even **below breakeven levels**. Let's see why... Second reason is events in Japan have **disrupted the supply chain** on both raw and finished materials.

Notes: This table reports firm name, earnings call date, and excerpts from earnings calls with high SCRisk and SCSentiment (both positive and negative) in Panels A and B, respectively.

Table IA.4 Components of Supply Chain Risk and Sentiment.

Topics	(1) SCRisk	(2) SCSentiment
Costs and commodity price risk	0.6848*** (0.0550)	-0.3608 (0.6276)
Technology and cyberattack risk	0.4951*** (0.0640)	1.7887*** (0.6423)
Climate risk and pandemics	0.9179*** (0.0714)	-1.0465 (0.7168)
Market uncertainty and regions	0.2035*** (0.0775)	-0.6449 (0.9152)
Liquidity	0.3843*** (0.0610)	-1.0770 (0.7366)
Analysts and financial issues	-0.0421 (0.0447)	-0.6539 (0.6154)
Acquisitions	0.2419*** (0.0600)	0.2764 (0.6579)
Firm FE	Y	Y
Industry x year FE	Y	Y
Observations	36,430	36,430
Adjusted R-squared	0.2166	0.3669

Notes: This table reports the components that contribute to SCRisk and SCSentiment using topic analysis. We calculate the probability of each topic and regress SCRisk and SCSentiment on the topics' probabilities in columns (1) and (2), respectively. The topic probabilities are standardized by subtracting from each topic probability the mean and dividing by the standard deviation for ease of comparison. All variables are defined in Appendix A. Robust standard errors clustered by firm are in parentheses. Statistical significance at the 1%, 5%, and 10% level is denoted by ***, **, and *, respectively.

Table IA.5 Industry level Supply Chain Exposure.

SIC2	Top 10 Industries	SIC2	Bottom 10 Industries
22	Textile Mill Products	21	Tobacco Products
25	Furniture & Fixtures	27	Printing & Publishing
33	Primary Metal Industries	41	Local & Interurban Passenger Transit
34	Fabricated Metal Products	48	Communications
35	Industrial Machinery & Equipment	54	Food Stores
36	Electronic & Other Electric Equipment	58	Eating & Drinking Places
37	Transportation Equipment	72	Personal Services
42	Trucking & Warehousing	78	Motion Pictures
50	Wholesale Trade – Durable Goods	79	Amusement & Recreation Services
75	Auto Repair, Services, & Parking	82	Educational Services

Notes: This table reports the top and bottom 10 industries in terms of our measure of overall supply chain risk, SCRisk. Industry-year average of firms' SCRisk is used to rank the industries.

Table IA.6 Input Specificity and Stage of Production.

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Input Specificity						
	Dependent var: SCRisk			Dependent var: SCSentiment		
Input specificity	0.0906*** (0.0149)	0.0849*** (0.0149)	0.1813*** (0.0352)	1.0751*** (0.2414)	0.9810*** (0.2345)	-0.5727 (0.5100)
Size		0.1508*** (0.0442)	0.0774* (0.0434)		3.1836*** (0.6899)	3.3766*** (0.6534)
Tobin's Q		-0.0965** (0.0387)	-0.1024*** (0.0378)		1.6880** (0.6870)	1.0282 (0.6523)
Cash holdings		-0.4989* (0.2837)	-0.8364*** (0.3169)		-27.3071*** (4.2297)	-12.8597*** (4.5775)
Cash flow		-0.0425 (0.2589)	1.1318*** (0.2707)		0.3712 (4.3723)	-8.2426* (4.4595)
Year FE			Y			Y
Industry FE			Y			Y
Observations	32,591	32,591	32,591	32,591	32,591	32,591
Adj. R-squared	0.0044	0.0067	0.0884	0.0046	0.0148	0.0803
Panel B: Stage of Production						
	Dependent var: SCRisk			Dependent var: SCSentiment		
Crude dummy	-1.5646*** (0.1842)	-1.9767*** (0.1971)	-1.8757*** (0.2017)	-33.4761*** (1.7554)	-41.7152*** (2.0832)	-42.2735*** (2.0791)
Size		0.1647*** (0.0459)	0.0602 (0.0467)		3.6366*** (0.6965)	3.2562*** (0.7235)
Tobin's Q		-0.1253*** (0.0401)	-0.2147*** (0.0404)		1.3403* (0.6868)	1.1454 (0.7059)
Cash holdings		-0.8478*** (0.2982)	-0.5889** (0.2968)		-32.1853*** (4.2742)	-32.6849*** (4.2958)
Cash flow		-0.1232 (0.2655)	1.1335*** (0.2847)		-2.9841 (4.3708)	-1.5845 (4.5351)
Year FE			Y			Y
Observations	30,686	30,686	30,686	32,591	32,591	32,591
Adj. R-squared	0.0012	0.0048	0.0648	0.0042	0.0174	0.0263

Notes: This table relate a firm's SCRisk to its input specificity and stage of production in Panels A and B, respectively. In Panel A, the main independent variable is *Input specificity*, which is constructed based on input-output tables and Rauch (1999) classification of differentiated products, and varies across industries and years. In Panel B, the main independent variable is *Crude dummy*, which is an indicator that equals one if the firm is producing raw commodities based on its three-digit SIC code, following the Stage of Production by the U.S. Bureau of Labor Statistics. Firm controls include size, Tobin's Q, cash holdings, and cash flow at $t-1$. All variables are defined in Appendix A. Robust standard errors clustered by firm are in parentheses. Statistical significance at the 1%, 5%, and 10% level is denoted by ***, **, and *, respectively.

Table IA.7 Variance Decomposition of SCRisk and SCSentiment.

	(1)	(2)	(3)	(4)
Panel A: SCRisk – 2-digit SIC				
Year FE	Y			
Industry FE		Y		
Industry x year FE			Y	Y
Firm FE				Y
Adjusted R-squared	0.0540	0.0261	0.1031	0.1924
R-squared	0.0545	0.0276	0.1292	0.3089
Panel B: SCSentiment – 2-digit SIC				
Year FE	Y			
Industry FE		Y		
Industry x year FE			Y	Y
Firm FE				Y
Adjusted R-squared	0.0091	0.0668	0.0836	0.3651
R-squared	0.0096	0.0683	0.1102	0.4567
Panel C: SCRisk – 3-digit SIC				
Year FE	Y			
Industry FE		Y		
Industry x year FE			Y	Y
Firm FE				Y
Adjusted R-squared	0.0540	0.0357	0.1188	0.2020
R-squared	0.0545	0.0414	0.1956	0.3648
Panel D: SCSentiment – 3-digit SIC				
Year FE	Y			
Industry FE		Y		
Industry x year FE			Y	Y
Firm FE				Y
Adjusted R-squared	0.0091	0.1061	0.1264	0.3839
R-squared	0.0096	0.1114	0.2026	0.5096
Panel E: SCRisk – 4-digit SIC				
Year FE	Y			
Industry FE		Y		
Industry x year FE			Y	Y
Firm FE				Y
Adjusted R-squared	0.0540	0.0421	0.1387	0.2167
R-squared	0.0545	0.0508	0.2469	0.4080
Panel F: SCSentiment – 4-digit SIC				
Year FE	Y			

Industry FE		Y		
Industry x year FE			Y	Y
Firm FE				Y
Adjusted R-squared	0.0091	0.1195	0.1440	0.3959
R-squared	0.0096	0.1276	0.2516	0.5434

Notes: This table reports adjusted R-squared and R-squared from the projection of SCRisk and SCSentiment on various sets of fixed effects, as indicated in the table. Industries are classified at the two-digit SIC code level in Panels A and B, three-digit SIC code level in Panels C and D, and four-digit SIC code level in Panels E and F.

Table IA.8 Supply Chain Risk, Firm Volatility, and Returns.

	(1)	(2)	(3)
Panel A: Dependent variable – realized volatility			
SCRisk	0.0029* (0.0016)	0.0038** (0.0016)	0.0030* (0.0015)
SCSentiment		-0.0008*** (0.0001)	-0.0008*** (0.0001)
Political Risk			0.0483*** (0.0145)
Climate Risk			0.0204 (0.0213)
Number of Firms	2,626	2,626	2,626
Number of Years	18	18	18
Panel B: Dependent variable – 30-day average abnormal return			
SCSentiment	0.0006*** (0.0002)	0.0006*** (0.0002)	0.0006*** (0.0002)
SCRisk		-0.0034 (0.0023)	-0.0037* (0.0020)
Political Risk			0.0121 (0.0333)
Climate Risk			0.0544 (0.0474)
Number of Firms	2,626	2,626	2,626
Number of Years	18	18	18

Notes: This table reports estimates of Fama-MacBeth regressions of firms' yearly realized volatility and 30-day average abnormal return on SCRisk and SCSentiment during the year in Panels A and B, respectively. The dependent variable in Panel A is *Realized Volatility*, computed as a firm's standard deviation of daily returns in that year. In Panel B, since earnings conference calls focus on the determinants of earnings and are therefore backward looking, we test how sentiment is associated with past stock returns. It is also plausible that the conversation is focused on the latest shocks. For this reason, in Panel B, the dependent variable is *30-day average abnormal return*, computed as the average market model abnormal stock returns for the 30 days prior to the earnings call date, which we average over the year. In panel B only, we multiply the dependent variable by a factor of 1,000 for presentation. Firm-level abnormal returns are obtained by estimating the market model over the [-255, -31] day interval. The Political risk measure is from Hassan et al. (2019). The Climate risk measure is from Sautner et al. (2023). All variables are defined in Appendix A. Statistical significance at the 1%, 5%, and 10% level is denoted by ***, **, and *, respectively.

Table IA.9 Firm Characteristics and Supply Chain Sentiment.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Dependent variable: SCSentiment									
Different continents	0.4154 (3.0913)	0.4155 (3.0913)	-4.1668 (3.0795)	-4.1853 (3.0760)	-5.2573* (3.0610)	-5.2549* (3.0609)	-5.4323* (3.0594)	-5.4795* (3.0626)	-5.1498* (3.0566)	-5.1878* (3.0543)
Relative size		0.0216 (0.0467)	-0.0580** (0.0282)	-0.0597** (0.0281)	-0.0568** (0.0276)	-0.0573** (0.0276)	-0.0776** (0.0314)	-0.0776** (0.0315)	-0.0764** (0.0308)	-0.0760** (0.0308)
Size			3.6822*** (0.5519)	3.5939*** (0.5907)	2.8492*** (0.6099)	2.8459*** (0.6116)	2.0328*** (0.6367)	2.1925*** (0.6469)	1.8977*** (0.6715)	2.0834*** (0.6675)
Average number of suppliers industry				0.4114 (1.2412)	0.7682 (1.2290)	0.7737 (1.2297)	-0.3689 (1.1869)	-0.3810 (1.1871)	-0.2391 (1.1874)	-0.3817 (1.1848)
Multinational					9.3985*** (2.2779)	9.4008*** (2.2788)	8.8948*** (2.2595)	8.8262*** (2.2625)	8.8118*** (2.2622)	8.9349*** (2.2691)
Customer dependence						-0.3828 (2.1377)	-0.2679 (2.1412)	-0.3381 (2.1444)	-0.5673 (2.1677)	-0.5136 (2.1653)
Market share							1.4184*** (0.4941)	1.4061*** (0.4924)	1.4417*** (0.4924)	1.4076*** (0.4904)
Financial constraint								1.4538 (1.9496)	1.8228 (1.9637)	1.9680 (1.9777)
Institutional ownership									4.9404* (2.9804)	4.3310 (2.8940)
Tobin's Q										1.0829 (0.6615)
Ind x year FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Observations	36,043	36,043	36,043	36,043	36,043	36,043	36,043	36,043	36,043	36,043
Adj. R-squared	0.0819	0.0819	0.0861	0.0861	0.0876	0.0876	0.0892	0.0892	0.0894	0.0897

Notes: This table relates SCSentiment to contemporaneous firm characteristics in an annual panel. The main independent variable in column (1) is *Different continents*, which is the fraction of a firm's suppliers who are located in a different continent over the total number of suppliers. The additional independent variable in column (2) is *Relative Size*, defined as a firm's total assets divided by the average total assets of its suppliers. The additional independent variable in column (3) is *Size*, defined as the natural logarithm of the firm's total assets. The additional independent variable in column (4) is *Average number of suppliers by industry*, defined as the average of a firm's number of suppliers within each of the three-digit SIC industries for which we observe suppliers from Factset Revere. The additional independent variable in column (5) is *Multinational*, defined as an indicator variable that equals one if the firm has geographical segments in other countries based on Compustat segment data. The additional independent variable in column (6) is *Customer dependence*, defined as an indicator variable that equals one if the firm reports major customers that purchase at least 10% of its sales in Compustat segment data. The additional independent variable in column (7) is

Market Share, defined as a firm's sales divided by the total sales in the firm's industry, which is scaled up by a factor of 100. The additional independent variable in column (8) is *Financial constraint*, which is an indicator variable that equals one if the Whited-Wu (2006) proxy for firm-level financial constraints is above the median of our sample. The additional independent variable in column (9) is *Institutional ownership*, which is the fraction of the firm's shares owned by financial institutions, which we obtain from 13F filings. The additional independent variable in column (10) is *Tobin's Q*, defined as assets minus cash and cash equivalent securities plus book value on equity scaled by assets. The unit of observation is a firm-year. Industries are classified at the two-digit SIC code level. All variables are defined in Appendix A. Robust standard errors clustered by firm are in parentheses. Statistical significance at the 1%, 5%, and 10% level is denoted by ***, **, and *, respectively.

Table IA.10 First-stage Regressions.

	(1) SCRisk	(2) SCRisk
Suppliers' SCRisk	0.0102*** (0.0037)	0.0100*** (0.0037)
SCSentiment		0.0020 (0.0012)
Size	0.0220 (0.1490)	0.0171 (0.1485)
Tobin's Q	-0.1180** (0.0523)	-0.1190** (0.0522)
Cash holdings	1.6313** (0.6413)	1.6269** (0.6413)
Cash flow	-0.2030 (0.4108)	-0.2131 (0.4105)
Firm FE	Y	Y
Industry x year FE	Y	Y
Observations	24,139	24,139
Adjusted R-squared	0.1222	0.1226
First-stage F-stat	13.384	12.916

Notes: This table reports estimates of the first-stage regressions. We regress the firm's SCRisk on its suppliers' SCRisk, which is our instrumental variable. The unit of observation in each regression is a firm-year. Firm controls include size, Tobin's Q, cash holdings, and cash flow. All variables are defined in Appendix A. Robust standard errors clustered by firm are in parentheses. Statistical significance at the 1%, 5%, and 10% level is denoted by ***, **, and *, respectively.

Table IA.11 Instrumenting for both SCRisk and SCSentiment.

Panel A: Firms' M&As			
	(1) M&A with supplier	(2) M&A with customer	(3) Unrelated M&As
SCRisk	0.0819** (0.0366)	0.0953** (0.0430)	-0.0070 (0.0129)
SCSentiment	-0.0024 (0.0039)	-0.0046 (0.0047)	-0.0011 (0.0014)
Size	-0.0169 (0.0173)	-0.0071 (0.0203)	0.0019 (0.0060)
Tobin's Q	0.0148* (0.0077)	0.0189** (0.0090)	0.0006 (0.0027)
Cash holdings	-0.0264 (0.0821)	-0.0727 (0.0966)	0.0643** (0.0289)
Cash flow	0.1426*** (0.0438)	0.1204** (0.0528)	0.0502*** (0.0145)
Firm FE	Y	Y	Y
Industry x year FE	Y	Y	Y
Observations	24,139	24,139	24,139
First-stage F-stat	2.628	2.628	2.628

Panel B: Supply chain composition					
	(1)	(2)	(3)	(4)	(5)
	Number of suppliers	Number of suppliers in the same continent	Number of U.S. suppliers	Number of suppliers in different continents	Number of industry leader suppliers
SCRisk	4.2452** (1.8066)	2.0202** (0.9127)	1.8566** (0.8728)	1.0215** (0.4626)	1.6626** (0.7073)
SCSentiment	-0.1618 (0.1918)	-0.1034 (0.0974)	-0.1076 (0.0930)	-0.0484 (0.0493)	-0.0575 (0.0747)
Size	2.5912*** (0.8612)	1.4177*** (0.4385)	1.3815*** (0.4233)	0.5305** (0.2198)	1.2514*** (0.3363)
Tobin's Q	0.6480* (0.3815)	0.2692 (0.1902)	0.2919 (0.1819)	0.1532 (0.0962)	0.2619* (0.1474)
Cash holdings	-6.2853 (4.0552)	-3.6985* (2.0396)	-3.3957* (1.9310)	-1.4821 (1.0177)	-3.1664** (1.5849)
Cash flow	-0.2442 (2.2133)	-0.2236 (1.1158)	-0.2319 (1.0573)	-0.0293 (0.5663)	-0.3546 (0.8523)
Firm FE	Y	Y	Y	Y	Y
Industry x year FE	Y	Y	Y	Y	Y
Observations	24,139	24,139	24,139	24,139	24,139
First-stage F-stat	2.628	2.628	2.628	2.628	2.628

Notes: This table reports instrumental variable estimates of the effects of SCRisk at $t-1$ on firms' M&As and number of suppliers at t in Panel A and Panel B, respectively. The dependent variables in columns (1) and (2) of Panel A are *M&A with supplier* and *M&A with customer*, which are indicator variables that equal one if the firm conducts an M&A with a firm from an upstream or downstream industry, respectively. The dependent variable in column (3) of Panel A is *Unrelated M&A*, which is an indicator variable that equals one if the firm conducts an M&A with a firm that is neither in an upstream nor a downstream industry. The dependent variables in columns (1) to (5) of Panel B are number of suppliers, number of suppliers in the same continent as the firm, number of U.S. suppliers, number of suppliers in different continents, and number of industry leader suppliers, respectively. The unit of observation in each regression is a firm-year. Firm controls include size, Tobin's Q, cash holdings, and cash flow at $t-1$. We instrument for firm's supply chain risk with the supply chain risk of its suppliers and firm's supply chain sentiment with the supply chain sentiment of its suppliers. All variables are defined in Appendix A. Robust standard errors clustered by firm are in parentheses. Statistical significance at the 1%, 5%, and 10% level is denoted by ***, **, and *, respectively.

Table IA.12 Supply Chain Risk and the Composition of Supply Chains.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Number of suppliers	Number of suppliers in the same continent	Number of U.S. suppliers	Number of suppliers in different continents	Number of industry leader suppliers	Number of suppliers	Number of suppliers in the same continent	Number of U.S. suppliers	Number of suppliers in different continents	Number of industry leader suppliers
	OLS					IV				
SCRisk	0.0193*** (0.0067)	0.0121*** (0.0040)	0.0099*** (0.0035)	0.0035 (0.0022)	0.0055* (0.0029)	2.8039*** (1.0466)	1.0988** (0.4575)	0.8981** (0.3813)	0.5907** (0.2356)	1.1505*** (0.4402)
Size	2.3159*** (0.2404)	1.2175*** (0.1192)	1.1648*** (0.1116)	0.4397*** (0.0593)	1.1591*** (0.1063)	2.2187*** (0.4801)	1.1796*** (0.1975)	1.1338*** (0.1679)	0.4192*** (0.1081)	1.1191*** (0.1953)
Tobin's Q	0.0624 (0.0676)	-0.0235 (0.0334)	0.0159 (0.0303)	0.0070 (0.0181)	0.0355 (0.0287)	0.3902* (0.2027)	0.1044 (0.0854)	0.1205* (0.0711)	0.0762* (0.0441)	0.1703** (0.0837)
Cash holdings	0.2401 (0.6253)	-0.6553* (0.3546)	-0.6243* (0.3345)	0.0695 (0.1742)	-0.5947** (0.2964)	-4.2908* (2.5828)	-2.4235** (1.0855)	-2.0694** (0.8971)	-0.8859 (0.5671)	-2.4579** (1.0755)
Cash flow	-1.9705*** (0.3657)	-1.1747*** (0.2191)	-1.1679*** (0.2034)	-0.4914*** (0.1122)	-1.0022*** (0.1714)	-1.3579 (1.2107)	-0.9356* (0.4926)	-0.9725** (0.4133)	-0.3622 (0.2709)	-0.7503 (0.4906)
Firm FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Ind x year FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Observations	24,139	24,139	24,139	24,139	24,139	24,139	24,139	24,139	24,139	24,139
Adj. R-squared	0.8268	0.7533	0.7624	0.6594	0.8317	-	-	-	-	-
First-stage coeff	-	-	-	-	-	0.0102*** (0.0037)	0.0102*** (0.0037)	0.0102*** (0.0037)	0.0102*** (0.0037)	0.0102*** (0.0037)
First-stage F-stat	-	-	-	-	-	13.384	13.384	13.384	13.384	13.384

Notes: This table reports estimates of the effects of a firm's SCRisk at $t-1$ on its number of suppliers at t . The dependent variables in columns (1) and (6), (2) and (7), (3) and (8), (4) and (9), and (5) and (10) are the number of suppliers, the number of suppliers in the same continent as the firm, the number of U.S. suppliers, the number of suppliers in different continents, and the number of suppliers that are industry leaders, respectively. Industry leaders are defined as suppliers with sales above the median of their 3-digit SIC industry. Firm controls include size, Tobin's Q, cash holdings, and cash flow at $t-1$. The unit of observation is a firm-year. Industries are classified at the two-digit SIC code level. Columns 1-5 report OLS regression results. Columns 6-10 use the supply chain risk of its suppliers as an instrument. All variables are defined in Appendix A. Robust standard errors clustered by firm are in parentheses. Statistical significance at the 1%, 5%, and 10% level is denoted by ***, **, and *, respectively.

Table IA.13 Supply Chain Risk and Vertical M&As.

	(1) M&A with supplier	(2) M&A with customer	(3) Unrelated M&As	(4) M&A with supplier	(5) M&A with customer	(6) Unrelated M&As
	OLS			IV		
SCRisk	0.0007** (0.0003)	0.0007** (0.0003)	-0.0002* (0.0001)	0.0603** (0.0260)	0.0542** (0.0235)	-0.0171* (0.0102)
Size	-0.0205*** (0.0062)	-0.0160*** (0.0049)	-0.0011 (0.0044)	-0.0226** (0.0110)	-0.0178* (0.0094)	-0.0005 (0.0049)
Tobin's Q	0.0039* (0.0022)	0.0053*** (0.0018)	0.0008 (0.0014)	0.0109** (0.0048)	0.0115*** (0.0043)	-0.0012 (0.0021)
Cash holdings	0.1001*** (0.0237)	0.0708*** (0.0199)	0.0512*** (0.0187)	0.0030 (0.0637)	-0.0162 (0.0565)	0.0786*** (0.0285)
Cash flow	0.1128*** (0.0141)	0.0770*** (0.0116)	0.0460*** (0.0094)	0.1260*** (0.0293)	0.0888*** (0.0259)	0.0423*** (0.0119)
Firm FE	Y	Y	Y	Y	Y	Y
Industry x year FE	Y	Y	Y	Y	Y	Y
Observations	24,139	24,139	24,139	24,139	24,139	24,139
Adjusted R-squared	0.1379	0.1178	0.1172	-	-	-
First-stage coeff	-	-	-	0.0102*** (0.0037)	0.0102*** (0.0037)	0.0102*** (0.0037)
First-stage F-stat	-	-	-	13.384	13.384	13.384

Notes: This table reports estimates of the effects of SCRisk at $t-1$ on the probability that a firm is involved in M&As at t . The dependent variables in columns (1) and (4) and (2) and (5) are *M&A with supplier* and *M&A with customer*, which are indicator variables that equal one if the firm conducts an M&A with a firm in an upstream or downstream industry, respectively. The dependent variable in columns (3) and (6) is *Unrelated M&A*, which is an indicator variable that equals one if the firm conducts an M&A with a firm that is not in an upstream, downstream, or the same industry. The unit of observation in each regression is a firm-year. Firm controls include size, Tobin's Q, cash holdings, and cash flow at $t-1$. Industries are classified at the two-digit SIC code level. Columns 1-3 report OLS regression results. Columns 4-6 use the supply chain risk of its suppliers as an instrument. All variables are defined in Appendix A. Robust standard errors clustered by firm are in parentheses. Statistical significance at the 1%, 5%, and 10% level is denoted by ***, **, and *, respectively.

Table IA.14 SCRisk and Vertical M&As – Alternative Definitions of Upstream and Downstream Industries.

	(1) M&A with supplier	(2) M&A with customer	(3) Unrelated M&As
SCRisk	0.0570** (0.0250)	0.0590** (0.0251)	-0.0173* (0.0105)
SCSentiment	-0.0001 (0.0001)	-0.0001 (0.0001)	0.0000 (0.0000)
Size	-0.0224** (0.0104)	-0.0127 (0.0098)	-0.0007 (0.0049)
Tobin's Q	0.0113** (0.0046)	0.0123*** (0.0045)	-0.0012 (0.0021)
Cash holdings	-0.0095 (0.0603)	-0.0453 (0.0597)	0.0785*** (0.0287)
Cash flow	0.0992*** (0.0274)	0.0598** (0.0271)	0.0422*** (0.0120)
Firm FE	Y	Y	Y
Industry x year FE	Y	Y	Y
Observations	24,139	24,139	24,139
First-stage coeff	0.0100*** (0.0037)	0.0100*** (0.0037)	0.0100*** (0.0037)
First-stage F-stat	12.916	12.916	12.916

Notes: This table reports estimates of the effects of SCRisk on the probability that a firm is involved in an M&A. The dependent variables in columns (1) and (2) are *M&A with supplier* and *M&A with customer*, which are indicator variables that equal one if the firm conducts an M&A with a firm from an upstream or downstream industry, respectively. A target firm is considered to be a supplier (customer) if the acquirer's industry purchases (sells) at least one percent of its inputs (outputs) from (to) the target's industry. The dependent variable in column (3) is *Unrelated M&A*, which is an indicator variable that equals one if the firm conducts an M&A with a firm that is not in an upstream, downstream, or the same industry. The unit of observation in each regression is a firm-year. Firm controls include size, Tobin's Q, cash holdings, and cash flow at *t-1*. We instrument for firm's supply chain risk with the supply chain risk of its suppliers. All variables are defined in Appendix A. Robust standard errors clustered by firm are in parentheses. Statistical significance at the 1%, 5%, and 10% level is denoted by ***, **, and *, respectively.

Table IA.15 Supply Chain Risk, Financial Constraints, and Vertical Integration.

	(1) M&A with supplier	(2) M&A with customer	(3) Unrelated M&As
Panel A: Hadlock-Pierce financial constraint measure			
SCRisk	0.0010*** (0.0004)	0.0010*** (0.0004)	-0.0003* (0.0002)
HP FC dummy	0.0029 (0.0105)	-0.0050 (0.0083)	-0.0127* (0.0076)
SCRisk x HP FC dummy	-0.0017** (0.0007)	-0.0014** (0.0007)	0.0003 (0.0003)
Firm controls	Y	Y	Y
Firm FE	Y	Y	Y
Industry x year FE	Y	Y	Y
Observations	24,139	24,139	24,139
Adjusted R-squared	0.1380	0.1179	0.1173
Panel B: Whited-Wu financial constraint measure			
SCRisk	0.0010*** (0.0004)	0.0010*** (0.0004)	-0.0002 (0.0002)
WW FC dummy	-0.0049 (0.0072)	-0.0072 (0.0063)	-0.0056 (0.0052)
SCRisk x WW FC dummy	-0.0016** (0.0008)	-0.0013* (0.0007)	-0.0001 (0.0004)
Firm controls	Y	Y	Y
Firm FE	Y	Y	Y
Industry x year FE	Y	Y	Y
Observations	24,139	24,139	24,139
Adjusted R-squared	0.1381	0.1180	0.1172

Notes: This table reports OLS estimates of the effects of SCRisk at $t-1$ on firms' M&As at t using different measures of firms' financial constraints. Panel A and B use the Hadlock and Pierce (2010) and Whited-Wu (2006) measures, respectively, to define financial constraints. The dependent variables in columns (1) and (2) are *M&A with supplier* and *M&A with customer*, which are indicator variables that equal one if the firm conducts an M&A with a firm from an upstream or downstream industry, respectively. The dependent variable in column (3) is *Unrelated M&A*, which is an indicator variable that equals one if the firm conducts an M&A with a firm that is not in an upstream, downstream, or the same industry. The unit of observation in each regression is a firm-year. Firm controls include SCSentiment, size, Tobin's Q, cash holdings, and cash flow at $t-1$. All variables are defined in Appendix A. Robust standard errors clustered by firm are in parentheses. Statistical significance at the 1%, 5%, and 10% level is denoted by ***, **, and *, respectively.

Table IA.16 Supply Chain Risk vs Political Risk and Climate Risk.

Panel A: Firms' M&As			
	(1) M&A with supplier	(2) M&A with customer	(3) Unrelated M&As
SCRisk	0.0613** (0.0268)	0.0549** (0.0242)	-0.0173* (0.0105)
SCSentiment	-0.0001 (0.0001)	-0.0001 (0.0001)	0.0000 (0.0000)
Size	-0.0220** (0.0111)	-0.0173* (0.0095)	-0.0007 (0.0049)
Tobin's Q	0.0111** (0.0049)	0.0117*** (0.0044)	-0.0012 (0.0021)
Cash holdings	0.0024 (0.0646)	-0.0165 (0.0572)	0.0785*** (0.0286)
Cash flow	0.1259*** (0.0295)	0.0886*** (0.0261)	0.0423*** (0.0120)
Political risk	-0.0033 (0.0084)	-0.0025 (0.0072)	-0.0001 (0.0025)
Climate risk	-0.0102 (0.0085)	-0.0102 (0.0076)	0.0011 (0.0034)
Firm FE	Y	Y	Y
Industry x year FE	Y	Y	Y
Observations	24,139	24,139	24,139
First-stage coeff	0.0100*** (0.0037)	0.0100*** (0.0037)	0.0100*** (0.0037)
First-stage F-stat	12.916	12.916	12.916

Panel B: Supply chain composition					
	(1)	(2)	(3)	(4)	(5)
	Number of suppliers	Number of suppliers in the same continent	Number of U.S. suppliers	Number of suppliers in different continents	Number of industry leader suppliers
SCRisk	2.8415*** (1.0773)	1.1142** (0.4699)	0.9102** (0.3914)	0.6002** (0.2427)	1.1650** (0.4527)
SCSentiment	-0.0048 (0.0041)	-0.0016 (0.0017)	-0.0012 (0.0014)	-0.0011 (0.0009)	-0.0018 (0.0017)
Size	2.2420*** (0.4827)	1.1885*** (0.1986)	1.1409*** (0.1686)	0.4241*** (0.1089)	1.1285*** (0.1961)
Tobin's Q	0.3958* (0.2060)	0.1073 (0.0868)	0.1228* (0.0723)	0.0777* (0.0449)	0.1725** (0.0851)
Cash holdings	-4.3186* (2.6233)	-2.4397** (1.1010)	-2.0830** (0.9091)	-0.8949 (0.5771)	-2.4680** (1.0910)
Cash flow	-1.3682 (1.2222)	-0.9461* (0.4973)	-0.9828** (0.4168)	-0.3628 (0.2742)	-0.7565 (0.4950)
Political risk	-0.0906 (0.3363)	-0.1587 (0.1180)	-0.1552 (0.0959)	-0.0283 (0.0652)	-0.0513 (0.1423)
Climate risk	-0.6371* (0.3475)	-0.2327 (0.1471)	-0.1846 (0.1223)	-0.1163 (0.0766)	-0.2682* (0.1451)
Firm FE	Y	Y	Y	Y	Y
Industry x year FE	Y	Y	Y	Y	Y
Observations	24,139	24,139	24,139	24,139	24,139
First-stage coeff	0.0100*** (0.0037)	0.0100*** (0.0037)	0.0100*** (0.0037)	0.0100*** (0.0037)	0.0100*** (0.0037)
First-stage F-stat	12.916	12.916	12.916	12.916	12.916

Notes: This table reports estimates of the effects of SCRisk at $t-1$ on firms' M&As and number of suppliers at t in Panel A and Panel B, respectively, controlling for two other sources of risk, political risk and climate risk. The Political risk measure is from Hassan et al. (2019). The Climate risk measure is from Sautner et al. (2023). The dependent variables in columns (1) and (2) of Panel A are *M&A with supplier* and *M&A with customer*, which are indicator variables that equal one if the firm conducts an M&A with a firm from an upstream or downstream industry, respectively. The dependent variable in column (3) of Panel A is *Unrelated M&A*, which is an indicator variable that equals one if the firm conducts an M&A with a firm that is not in an upstream, downstream, or the same industry. The dependent variables in columns (1) to (5) of Panel B are number of suppliers, number of suppliers in the same continent as the firm, number of U.S. suppliers, number of suppliers in different continents, and number of industry leader suppliers, respectively. The unit of observation in each regression is a firm-year. Firm controls include size, Tobin's Q, cash holdings, and cash flow at $t-1$. We instrument for firm's supply chain risk with the supply chain risk of its suppliers. All variables are defined in Appendix A. Robust standard errors clustered by firm are in parentheses. Statistical significance at the 1%, 5%, and 10% level is denoted by ***, **, and *, respectively.

Table IA.17 Supply Chain Risk vs Overall Sentiment.

Panel A: Firms' M&As			
	(1) M&A with supplier	(2) M&A with customer	(3) Unrelated M&As
SCRisk	0.0611** (0.0267)	0.0548** (0.0241)	-0.0173* (0.0104)
SCSentiment	-0.0001 (0.0001)	-0.0001 (0.0001)	0.0000 (0.0000)
Size	-0.0182 (0.0111)	-0.0146 (0.0095)	-0.0008 (0.0050)
Tobin's Q	0.0122** (0.0051)	0.0125*** (0.0045)	-0.0012 (0.0022)
Cash holdings	0.0064 (0.0640)	-0.0138 (0.0567)	0.0784*** (0.0285)
Cash flow	0.1251*** (0.0295)	0.0882*** (0.0262)	0.0423*** (0.0120)
Overall sentiment	0.2099*** (0.0718)	0.1553** (0.0650)	-0.0051 (0.0292)
Firm FE	Y	Y	Y
Industry x year FE	Y	Y	Y
Observations	24,139	24,139	24,139
First-stage coeff	0.0100*** (0.0037)	0.0100*** (0.0037)	0.0100*** (0.0037)
First-stage F-stat	12.916	12.916	12.916

Panel B: Supply chain composition					
	(1)	(2)	(3)	(4)	(5)
	Number of suppliers	Number of suppliers in the same continent	Number of U.S. suppliers	Number of suppliers in different continents	Number of industry leader suppliers
SCRisk	2.8408*** (1.0762)	1.1112** (0.4683)	0.9073** (0.3901)	0.6003** (0.2425)	1.1646** (0.4522)
SCSentiment	-0.0050 (0.0042)	-0.0017 (0.0017)	-0.0012 (0.0014)	-0.0011 (0.0009)	-0.0019 (0.0017)
Size	2.3108*** (0.4865)	1.2106*** (0.1995)	1.1549*** (0.1690)	0.4285*** (0.1098)	1.1537*** (0.1972)
Tobin's Q	0.4206** (0.2137)	0.1146 (0.0899)	0.1275* (0.0749)	0.0798* (0.0466)	0.1818** (0.0883)
Cash holdings	-4.2613 (2.6054)	-2.4134** (1.0907)	-2.0639** (0.9006)	-0.8926 (0.5730)	-2.4471** (1.0837)
Cash flow	-1.3561 (1.2263)	-0.9351* (0.4979)	-0.9714** (0.4172)	-0.3570 (0.2748)	-0.7494 (0.4969)
Overall sentiment	4.2728 (3.0231)	1.4420 (1.2638)	0.9675 (1.0626)	0.3513 (0.6752)	1.6053 (1.2644)
Firm FE	Y	Y	Y	Y	Y
Industry x year FE	Y	Y	Y	Y	Y
Observations	24,139	24,139	24,139	24,139	24,139
First-stage coeff	0.0100*** (0.0037)	0.0100*** (0.0037)	0.0100*** (0.0037)	0.0100*** (0.0037)	0.0100*** (0.0037)
First-stage F-stat	12.916	12.916	12.916	12.916	12.916

Notes: This table reports instrumental variable estimates of the effects of SCRisk at $t-1$ on firms' M&As and number of suppliers at t in Panel A and Panel B, respectively, controlling for the overall sentiment of a firm's earnings calls during year $t-1$. Overall sentiment is constructed by calculating the net sentiment in the whole earnings call transcript. The dependent variables in columns (1) and (2) of Panel A are *M&A with supplier* and *M&A with customer*, which are indicator variables that equal one if the firm conducts an M&A with a firm from an upstream or downstream industry, respectively. The dependent variable in column (3) of Panel A is *Unrelated M&A*, which is an indicator variable that equals one if the firm conducts an M&A with a firm that is not in an upstream, downstream, or the same industry. The dependent variables in columns (1) to (5) of Panel B are number of suppliers, number of suppliers in the same continent as the firm, number of U.S. suppliers, number of suppliers in different continents, and number of industry leader suppliers, respectively. The unit of observation in each regression is a firm-year. Firm controls include size, Tobin's Q, cash holdings, and cash flow at $t-1$. We instrument for firm's supply chain risk with the supply chain risk of its suppliers. All variables are defined in Appendix A. Robust standard errors clustered by firm are in parentheses. Statistical significance at the 1%, 5%, and 10% level is denoted by ***, **, and *, respectively.

Table IA.18 Supply Chain Risk – Replacing Potential Noise with Industry Median.

Panel A: Firms' M&As			
	(1) M&A with supplier	(2) M&A with customer	(3) Unrelated M&As
SCRisk	0.0643** (0.0290)	0.0577** (0.0262)	-0.0182 (0.0112)
SCSentiment	-0.0001 (0.0001)	-0.0001 (0.0001)	0.0000 (0.0000)
Size	-0.0236** (0.0115)	-0.0188* (0.0099)	-0.0003 (0.0050)
Tobin's Q	0.0110** (0.0051)	0.0116*** (0.0045)	-0.0012 (0.0022)
Cash holdings	-0.0030 (0.0684)	-0.0215 (0.0607)	0.0800*** (0.0297)
Cash flow	0.1288*** (0.0309)	0.0912*** (0.0274)	0.0416*** (0.0123)
Firm FE	Y	Y	Y
Industry x year FE	Y	Y	Y
Observations	24,139	24,139	24,139
First-stage coeff	0.0095*** (0.0037)	0.0095*** (0.0037)	0.0095*** (0.0037)
First-stage F-stat	11.804	11.804	11.804

Panel B: Supply chain composition					
	(1)	(2)	(3)	(4)	(5)
	Number of suppliers	Number of suppliers in the same continent	Number of U.S. suppliers	Number of suppliers in different continents	Number of industry leader suppliers
SCRisk	2.9856** (1.1849)	1.1675** (0.5108)	0.9530** (0.4247)	0.6302** (0.2649)	1.2238** (0.4969)
SCSentiment	-0.0050 (0.0043)	-0.0017 (0.0018)	-0.0013 (0.0015)	-0.0012 (0.0010)	-0.0019 (0.0018)
Size	2.1627*** (0.5041)	1.1571*** (0.2060)	1.1152*** (0.1748)	0.4077*** (0.1132)	1.0958*** (0.2049)
Tobin's Q	0.3937* (0.2144)	0.1053 (0.0896)	0.1210 (0.0745)	0.0771* (0.0465)	0.1715* (0.0885)
Cash holdings	-4.5854 (2.8100)	-2.5354** (1.1718)	-2.1592** (0.9656)	-0.9496 (0.6153)	-2.5769** (1.1674)
Cash flow	-1.2257 (1.2904)	-0.8857* (0.5212)	-0.9326** (0.4362)	-0.3334 (0.2871)	-0.6970 (0.5224)
Firm FE	Y	Y	Y	Y	Y
Industry x year FE	Y	Y	Y	Y	Y
Observations	24,139	24,139	24,139	24,139	24,139
First-stage coeff	0.0095*** (0.0037)	0.0095*** (0.0037)	0.0095*** (0.0037)	0.0095*** (0.0037)	0.0095*** (0.0037)
First-stage F-stat	11.804	11.804	11.804	11.804	11.804

Notes: This table reports estimates of the effects of SCRisk at $t-1$ on firms' M&As and number of suppliers at t in Panel A and Panel B, respectively. We replace firm-year observations of SCRisk in which more than 75% of the snippets that incorporate supply chain risk discussions are associated with probability of general financial or analyst and liquidity issues in the top quartile with the 2-digit SIC industry median of SCRisk. The dependent variables in columns (1) and (2) of Panel A are *M&A with supplier* and *M&A with customer*, which are indicator variables that equal one if the firm conducts an M&A with a firm from an upstream or downstream industry, respectively. The dependent variable in column (3) of Panel A is *Unrelated M&A*, which is an indicator variable that equals one if the firm conducts an M&A with a firm that is not in an upstream, downstream, or the same industry. The dependent variables in columns (1) to (5) of Panel B are number of suppliers, number of suppliers in the same continent as the firm, number of U.S. suppliers, number of suppliers in different continents, and number of industry leader suppliers, respectively. The unit of observation in each regression is a firm-year. Firm controls include size, Tobin's Q, cash holdings, and cash flow at $t-1$. We instrument for firm's supply chain risk with the supply chain risk of its suppliers. All variables are defined in Appendix A. Robust standard errors clustered by firm are in parentheses. Statistical significance at the 1%, 5%, and 10% level is denoted by ***, **, and *, respectively.

Table IA.19 Supply Chain Risk – Omitting Potential Noise.

Panel A: Firms' M&As			
	(1) M&A with supplier	(2) M&A with customer	(3) Unrelated M&As
SCRisk	0.0657** (0.0310)	0.0595** (0.0281)	-0.0177 (0.0116)
SCSentiment	-0.0001 (0.0001)	-0.0001 (0.0001)	0.0000 (0.0000)
Size	-0.0232* (0.0119)	-0.0176* (0.0103)	-0.0005 (0.0051)
Tobin's Q	0.0118** (0.0055)	0.0124** (0.0049)	-0.0012 (0.0023)
Cash holdings	-0.0077 (0.0734)	-0.0278 (0.0656)	0.0772** (0.0309)
Cash flow	0.1304*** (0.0324)	0.0918*** (0.0290)	0.0424*** (0.0126)
Firm FE	Y	Y	Y
Industry x year FE	Y	Y	Y
Observations	23,519	23,519	23,519
First-stage coeff	0.0093** (0.0038)	0.0093** (0.0038)	0.0093** (0.0038)
First-stage F-stat	10.684	10.684	10.684

Panel B: Supply chain composition					
	(1)	(2)	(3)	(4)	(5)
	Number of suppliers	Number of suppliers in the same continent	Number of U.S. suppliers	Number of suppliers in different continents	Number of industry leader suppliers
SCRisk	2.9100** (1.2023)	1.1121** (0.5133)	0.8988** (0.4239)	0.6323** (0.2772)	1.1599** (0.4914)
SCSentiment	-0.0051 (0.0044)	-0.0017 (0.0018)	-0.0012 (0.0015)	-0.0013 (0.0010)	-0.0019 (0.0018)
Size	2.1933*** (0.5043)	1.1742*** (0.2027)	1.1288*** (0.1710)	0.4145*** (0.1156)	1.1155*** (0.2001)
Tobin's Q	0.4058* (0.2225)	0.1093 (0.0917)	0.1224 (0.0757)	0.0823* (0.0495)	0.1727* (0.0897)
Cash holdings	-4.5736 (2.8796)	-2.5843** (1.1845)	-2.1978** (0.9692)	-1.0188 (0.6475)	-2.5259** (1.1669)
Cash flow	-1.1998 (1.3059)	-0.8765* (0.5169)	-0.9244** (0.4301)	-0.3360 (0.2977)	-0.6849 (0.5157)
Firm FE	Y	Y	Y	Y	Y
Industry x year FE	Y	Y	Y	Y	Y
Observations	23,519	23,519	23,519	23,519	23,519
First-stage coeff	0.0093** (0.0038)	0.0093** (0.0038)	0.0093** (0.0038)	0.0093** (0.0038)	0.0093** (0.0038)
First-stage F-stat	10.684	10.684	10.684	10.684	10.684

Notes: This table reports estimates of the effects of SCRisk at $t-1$ on firms' M&As and number of suppliers at t in Panel A and Panel B, respectively. We omit firm-year observations in which more than 75% of the snippets that incorporate supply chain risk discussions are associated with probability of general financial or analyst and liquidity issues in the top quartile. The dependent variables in columns (1) and (2) of Panel A are *M&A with supplier* and *M&A with customer*, which are indicator variables that equal one if the firm conducts an M&A with a firm from an upstream or downstream industry, respectively. The dependent variable in column (3) of Panel A is *Unrelated M&A*, which is an indicator variable that equals one if the firm conducts an M&A with a firm that is not in an upstream, downstream, or the same industry. The dependent variables in columns (1) to (5) of Panel B are number of suppliers, number of suppliers in the same continent as the firm, number of U.S. suppliers, number of suppliers in different continents, and number of industry leader suppliers, respectively. The unit of observation in each regression is a firm-year. Firm controls include size, Tobin's Q, cash holdings, and cash flow at $t-1$. We instrument for firm's supply chain risk with the supply chain risk of its suppliers. All variables are defined in Appendix A. Robust standard errors clustered by firm are in parentheses. Statistical significance at the 1%, 5%, and 10% level is denoted by ***, **, and *, respectively.